

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology
Practical - 26

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Class: BSC IT	Batch:-----
Date of Assignment:1/9/2026	Date/Time of Submission:1/9/2026

Aim: Perform logistic regression using Breeze. Classify a dataset with binary labels.

Code:

```
build.sbt  Demo26.scala x
1  import breeze.linalg._
2
3  ▶ object Demo26 extends App {
4
5      val x = DenseVector(1.0, 2.0, 3.0, 4.0, 5.0) // Dataset
6
7      val y = DenseVector(0.0, 0.0, 0.0, 1.0, 1.0) // Binary labels
8
9      // Initial weight and bias
10     var weight = 0.0
11     var bias = 0.0
12
13     // Sigmoid function
14     def sigmoid(z: Double): Double = {
15         1.0 / (1.0 + math.exp(-z))
16     }
17
18     // Train the model
19     for (i <- 1 to 1000) {
20         for (j <- 0 until x.length) {
21             val prediction = sigmoid(weight * x(j) + bias)
22             val error = prediction - y(j)
23             weight = weight - 0.01 * error * x(j)
24             bias = bias - 0.01 * error
25         }
26     }
```

```

27
28     // Classify the data
29     println("Predictions:")
30     for (i <- 0 until x.length) {
31         val probability = sigmoid(weight * x(i) + bias)
32         val result =
33             if (probability >= 0.5) 1 else 0
34         println("Input = " + x(i) + ", Probability = " + probability + ", Class = " + result)
35     }
36     println("\nAbhishek_S046")
37 }

```

Output:

```

Run Demo26 x
Predictions:
Input = 1.0, Probability = 0.0515501057576029, Class = 0
Input = 2.0, Probability = 0.16412826954644827, Class = 0
Input = 3.0, Probability = 0.4149888799701166, Class = 0
Input = 4.0, Probability = 0.7193154698480789, Class = 1
Input = 5.0, Probability = 0.9025176083649726, Class = 1

Abhishek_S046

Process finished with exit code 0

```

